

Midterm 2

Maze Solving

Material Available

- 1) `open_camera.py` -> Code to open your camera and check the video frame.
- 2) `robot_motion.py` -> Code to move and get position from robot.
- 3) `camera_utilities.py` -> Camera Helper function.
- 4) `robot_utilities.py` -> Robot Helper function.
- 5) `get_pixel_coordinates.py` -> Code to get pixel coordinates from camera window.
- 6) `Affine_transform.py` -> Code to convert the pixel coordinates to robot coordinates.
- 7) `Maze_Motion.py` -> Code to test your calibration, using a test maze path.

Camera Calibration

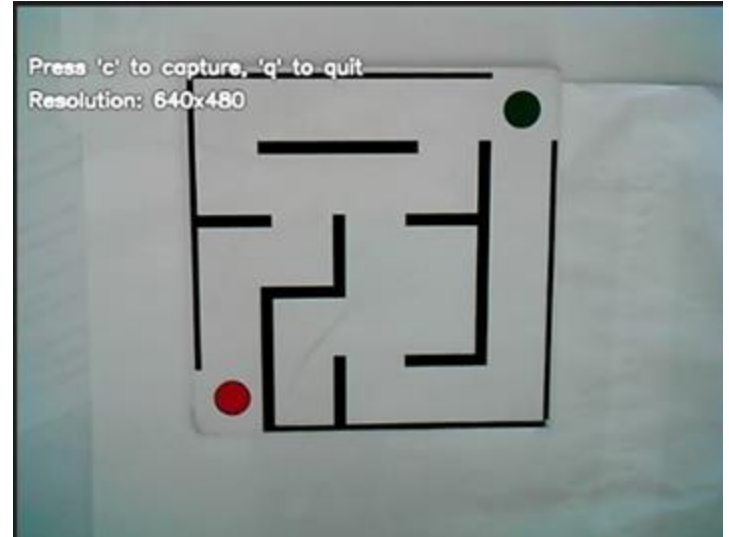
Understanding your frame

Step 1: Open your robot_motion.py file, and move your robot to home position. Make sure to change the port number to your robot.

Step2: Open your open_camera.py file file, and run the program to get the field of view of the camera, this will be your workspace. Make sure to change your camera port number according to your camera.

Note: Not all of the workspace in the camera field of view is accessible by the robot,

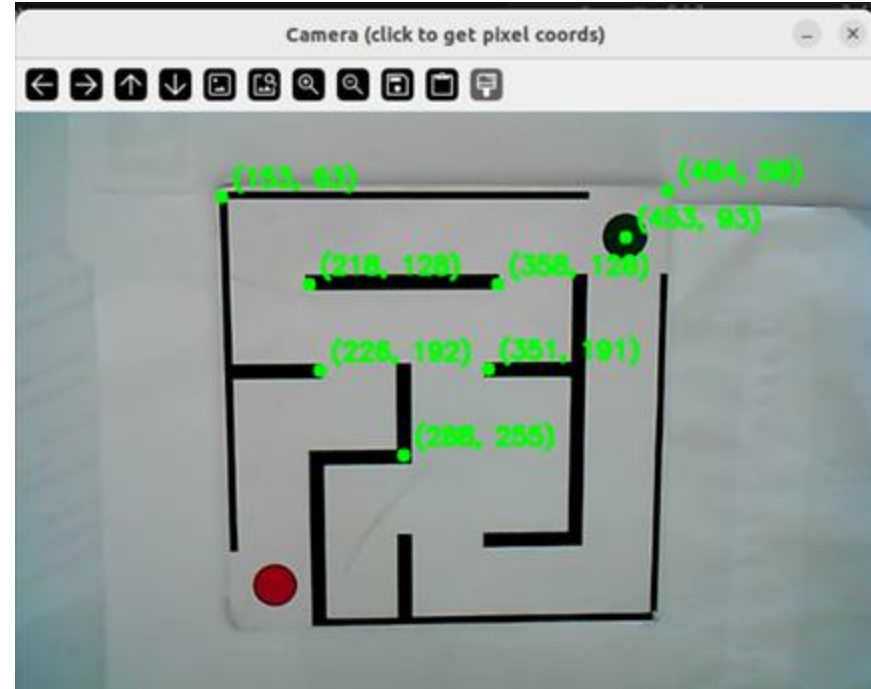
You must define your robot workspace and perform your calibration and maze solving in that region only.



Getting Pixel Points

Open the `get_pixel_coordinates.py` file and select the pixels points. Note them down which is displaced in the terminal.

```
rajesh@rajesh:~/My_Work/PhD_FALL_2025/My_Learnings/Perception/Maze_Solving/mid_term$ /usr/bin/python3
ork/PhD_FALL_2025/My_Learnings/Perception/Maze_Solving/final_code_mid_term_2/get_pixel_coordinates.py
qt.qpa.plugin: Could not find the Qt platform plugin "wayland" in "/home/rajesh/.local/lib/python3.
/qt/plugins"
Instructions:
  • Left-click on the video to print/overlay pixel coordinates (x, y)
  • Right-click to clear overlays
  • Press 'q' to quit
Clicked at: x=153, y=63
Clicked at: x=484, y=58
Clicked at: x=453, y=93
Clicked at: x=218, y=128
Clicked at: x=358, y=128
Clicked at: x=226, y=192
Clicked at: x=351, y=191
Clicked at: x=288, y=255
```



Get Robot Coordinates

Open the robot_motion.py file and use the `get_current_pose` function to get the robot position at the same coordinates. Ensure the points match when taking coordinates.

And updates these values in the affine_transform.py file as shown.

After running this file you will get a 2x3 affine matrix. Copy this matrix into the maze_motion.py file.

```
48 # move_to_specific_position(x=327, y=62, z=-45)
49
50 get_current_pose(device)
51 # move_to_specific_position(x=218, y=-41, z=-45)
52 # move_to_specific_position(x=245, y=-53, z=-45)
53
54
55 # current_pose, current_joint = get_current_pose()
56 # print(f"Final pose: {current_pose}")
57
58 # Close the device connection
```

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS

```
* rajesh@rajesh:~/My_work/PhD_FALL_2025/My_Learnings/Perception/Maze_Solving/mid_term$ /usr/bin/python3 /home/rajesh/My_Wor
k/PhD_FALL_2025/My_Learnings/Perception/Maze_Solving/final_code_mid_term_2/robot_motion.py
Homing the robot...
pose: Position(x=253.78980647460938, y=5.789537982940674, z=117.5219497680664, r=0.0), j: Joints(j1=1.21694816456604, j2=
6.178558349609375, j3=11.819925483783613, j4=-1.21694816456604)
current_pose
pose: Position
Focus folder in explorer (ctrl + click) 28209627783575e-10, z=150.4169921875, r=0.0)
* rajesh@rajesh:~/My_work/PhD_FALL_2025/My_Learnings/Perception/Maze_Solving/mid_term$
```

```
img_pts = np.array([
    [152, 63],
    [484, 58],
    [453, 93],
    [218, 128],
    [358, 128],
    [226, 192],
    [351, 191],
    [288, 255],
], dtype=np.float64)

rob_xy = np.array([
    [332.8875427246094, 50.575565338134766],
    [331.2615661621094, -94.13094329833984],
    [308.04815673828125, 27.38081169128418],
    [307.2852783203125, -34.91622543334961],
    [278.18890380859375, 24.748804092407227],
    [277.52886962890625, -31.535066604614258],
    [248.09780883789062, -3.083724021911621]
], dtype=np.float64)
```

Test your Maze Solution

For testing purposes, you can use the `get_pixel_coordinates.py` file, to simulate the maze path.

```
def main():
    device = pydobot.Dobot(port="/dev/ttyACM0")
    device.speed(50, 50)
    move_to_home(device)

    # Example pixel coordinates from clicks
    pixel_coords = [
        (462, 78),
        (398, 77),
        (395, 145),
        (271, 142),
        (270, 206),
        (201, 210),
        (204, 328),
    ]

    for (u, v) in pixel_coords:
        move_robot_point(device, M, u, v)
```

